

# Plexus Atlas — CompuCell3D

The second repository, and the first test of whether any of this transfers

1 August 2026

The twin of `atlas_jax_morph/atlas_note.pdf`. That one asked whether Plexus can say what one framework says. This one asks whether the *procedure* is a procedure.

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## 1. Why a second repository, and why this one

One repository cannot show saturation. The measurement `plexus2.tex` App. E.1 asks for — *does decomposing a broad collection of frameworks converge on a compact reusable vocabulary?* — is a claim about what happens *across* frameworks. With  $n = 1$  the curve is a shape, not an argument.

**Target: CompuCell3D (Cellular Potts).** Nominated by the `jax-morph` note before any of this work, on the grounds that it is *the biggest step away from everything we have*: a cell is not a point with per-cell state, it is a **set of lattice sites**. Its position is a derived centroid. Its dynamics are Metropolis flips of site ownership. There is no per-cell equation to subtract.

That is the point. Every measurement in the `jax-morph` atlas rested on cells being points and a differential test being a subtraction. **If the procedure is general it survives this; if it is secretly a particle-framework procedure, this is where it breaks.**

The prediction — made before running anything, and the reason this note exists — is in `TRANSFER.md`.

## 2. What transferred, measured rather than asserted

`STATUS.md` in the `jax-morph` atlas claimed ~60% of the Python was instrument and would copy as-is. Forking it is the test of that claim, and it came out slightly better than claimed.

| file                              | edits           | why                                                                                                                                                                                                                        |
|-----------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| record.py                         | none            | schema, ladder and the twelve rules are about the record, not the target. Self-test passes unchanged.                                                                                                                      |
| saturation.py                     | none            | to <i>transfer</i> : the ledger scores against the frozen baseline and nothing in it knows what a cell is. It was later <i>extended</i> — see §6 — but that is a defect the second repository exposed, not a porting cost. |
| atlas.py                          | none            | blast radius, rung, commit-or-revert — all HERE-relative.                                                                                                                                                                  |
| registry_view.py                  | none            | reads <code>plexus.operators</code> . Produced the same 52-contract baseline from the new folder.                                                                                                                          |
| agents/<br>run_spec.py            | none<br>5 lines | the six roles describe a job, not a repository.<br>output paths only ( <code>log/atlas_cc3d/</code> , <code>graphs_data/atlas_cc3d/</code> ).                                                                              |
| oracle.py, inventory.py, paper.py | rewrite         | target-specific by construction: the reference, its mechanism scan, its PDF.                                                                                                                                               |

Five of the seven instruments needed zero edits. That is the strongest evidence so far that the apparatus is a method rather than a one-off, and it cost about ten minutes to establish.

### 3. The oracle — where the risk actually is

TRANSFER.md predicted the oracle would be the largest risk, and a *target* risk rather than a *procedure* risk. It was both, and **it now passes**.

| gate                          | state |                                                                                                                                        |
|-------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| installs                      | done  | CompuCell3D 4.10.0 from its own conda channel into an isolated environment ( <code>/workspace/.conda_envs/cc3d-oracle</code> ), py312. |
| imports                       | done  | after supplying <code>libGL</code> — the container is headless and CompuCell3D links its graphics stack unconditionally.               |
| model definable in Python     | done  | <code>cc3d.core.PyCoreSpecs</code> , so the model is written in Python and never hand-typed as XML.                                    |
| isolation                     | done  | the Plexus interpreter <i>cannot import cc3d</i> . Checked, not assumed: a process that can reach the reference can borrow its answer. |
| runs headless                 | done  | 9 cells at 30 <sup>2</sup> , 45 at 60 <sup>2</sup> ; artefacts with provenance in <code>_oracle/runs/smoke/</code> .                   |
| deterministic at a fixed seed | done  | two runs at seed 42 are <i>bit-identical</i> across id, type, volume, surface and centre of mass; seed 7 differs.                      |

Getting there meant routing around three defects in CompuCell3D 4.10.0, none of them configuration:

1. The advertised pure-Python route (`PyCoreSpecs` → `service_cc3d`) passes the *list of specs* where a simulation file is expected: `get_custom_settings_path()` does `Path(<list>)`. Shim-ming that only exposes the next one — `CC3DSimService._run` asserts `os.path.isfile(<list>)`. **That route simply does not work in this release.**

2. CC3D **execs** the project's Python script with its own globals, so a steppable class defined in the same file raises **NameError: SteppableBasePy is not defined** at registration. The class must live in a separate importable module.
3. The output directory may not sit inside the project folder.

**The way through** is to use **PyCoreSpecs** for what it is genuinely good at — generating correct CC3DML, `<RandomSeed>` included — write a real `.cc3d` project, and run it through `cc3d/run_script.py`, the officially supported headless entry point, which never touches the broken path. The model stays defined in Python; only the transport is a file.

**Why this mattered enough to spend the time on.** `jax-morph` was chosen *because* it ran, and the first note said so in its own Phase 0. A target that cannot be run reduces the atlas to the Okuda situation — wrong operator, wrong parameter or wrong reading of a figure, with no way to tell which. **Determinism is the gate behind the gate:** without it, every later disagreement is unattributable between their noise and our error. Both now hold, so Phase 1 is unblocked.

## 4. The phases

Seven stages, the same ladder as the first atlas. **I stop at the end of each one and wait for you.** What differs is not the procedure but what each stage has to mean when a cell is a region of a lattice rather than a point.

### Phase 0 — Instruments and the oracle

DONE

**Goal.** Make every later measurement trustworthy before taking one.

The instruments came across for free (§2). **The oracle was the phase**, and it is where the risk always was: install, import, isolate, run headless, and be deterministic at a fixed seed.

**All six gates now pass** (§3), including the stop condition — the authors' own cell-sorting model runs twice from one seed and is bit-identical. Until that held, a later disagreement would have been unattributable between their noise and our error. It holds.

## Phase 0 — what actually happened

DONE

|                 | state | the issue found, and what was interesting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| instruments     | done  | Five of seven needed <b>zero edits</b> — <code>record.py</code> , <code>saturation.py</code> , <code>atlas.py</code> , <code>registry_view.py</code> , <code>agents/</code> . Both self-tests pass from the new folder and the same 52-contract baseline is reproduced, so the two campaigns are scored against one ruler.                                                                                                                                                                                                                                                                                                                                    |
| oracle installs | done  | CompuCell3D 4.10.0 from its own conda channel, isolated env, py312.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| oracle imports  | done  | needed <code>libGL</code> : the container is headless and CompuCell3D links its graphics stack unconditionally.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| isolation       | done  | the Plexus interpreter cannot <code>import cc3d</code> . Checked, not assumed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| headless run    | done  | <b>Three upstream defects had to be routed around.</b> The advertised pure-Python route ( <code>PyCoreSpecs</code> → <code>service_cc3d</code> ) passes the specs <i>list</i> where a file is expected — <code>Path(&lt;list&gt;)</code> , then <code>os.path.isfile(&lt;list&gt;)</code> . CC3D <code>execs</code> the project script with its own globals, so a steppable class in that file raises <code>NameError</code> at registration. And the output directory may not sit inside the project. The way through: generate CC3DML <i>from</i> <code>PyCoreSpecs</code> , write a real <code>.cc3d</code> project, run <code>cc3d/run_script.py</code> . |
| determinism     | done  | two runs at seed 42 bit-identical across id, type, volume, surface and centre of mass; seed 7 differs. <b>The first attempt compared a dump whose <code>xCOM/yCOM</code> were all zero</b> because <code>CenterOfMassPlugin</code> was not enabled — a weaker test than it looked, so the plugin was added and the check re-run.                                                                                                                                                                                                                                                                                                                              |

**And a diagnostic mistake worth recording.** Two runs appeared to produce *no output* and were reported as hangs at import. They were not: piping an unbuffered process through `grep` buffers its output, which is lost when the process is killed on timeout. The marks existed; the pipe ate them.

## Phase 1 — Read every mechanism

DONE

The excavator on each mechanism: what it does to the state, its equations, the role of every knob, and the **surprises** a reimplementer gets wrong.

**What is different here.** `jax-morph`'s mechanisms were 20 classes plus 4 architectural contracts a scan could not see. CompuCell3D's are *plugins* and *steppables* declared through XML or `PyCoreSpecs` — a different extraction problem, and one where the scan-versus-hand split will fall in a different place. **The 4 hand-added entries were among the most interesting things the first atlas found**; expecting an automatic scan to be sufficient here would be repeating a mistake we already know about.

The inventory is complete and it is bigger than the first atlas's: 35 mechanisms, against jax-morph's 24. Thirty come from `cc3d.core.PyCoreSpecs` — the framework's *own* registry of what a model can switch on, which is a better source than a syntax-tree scan because it is the project's index rather than our reading of it.

Five were added by hand, and they are the interesting ones. The first atlas found that 4 of its 24 were not classes; here the same is true and more so, because a Cellular Potts model's defining commitments are not plugins:

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|                                     |                                                                                                                                                                             |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>cell_as_lattice_domain</code> | a cell is a set of lattice sites sharing an id; volume is a count, position a derived centroid                                                                              |
| <code>metropolis_acceptance</code>  | $P = 1$ if $\Delta E \leq 0$ else $e^{-\Delta E/T}$ — discrete, stochastic, and with no pathwise derivative                                                                 |
| <code>energy_sum_composition</code> | mechanisms compose by <i>adding energy terms</i> , not by operator splitting. The direct counterpart of jax-morph's Lie–Trotter split, and the contrast is itself a finding |
| <code>mcs_time_unit</code>          | one MCS is one attempted copy per site. <b>There is no dt</b>                                                                                                               |
| <code>pixel_neighbourhood</code>    | the model's relation is a lattice adjacency, not an edge set built from positions                                                                                           |

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All 35 are now at *inspected*: 0 candidates, 0 validator violations, 0 blocked. The excavator read each mechanism at source in the installed package — 25 agent calls at roughly 4 minutes each, none reverted for a bad verdict.

The prompts had to be retargeted first, and that is the part worth recording. The forked `agents/` still described jax-morph: its paper, its extracted text, its guides directory. Running it unchanged would have sent 24 agents looking for `papers/jax-morph`. The preamble now describes a Cellular Potts framework, says where the source actually is (an absolute `file:Lnn` into the installed package), states plainly that **we have no extracted paper text** — “do not cite page numbers you have not read” — and adds a rule the first campaign never needed: *most CC3D mechanisms are energy terms, not updates; say so in `state_io` rather than forcing them into read/write language that does not fit*.

The record is checkable. `R3_code_path` resolves for every entry because each carries a real `file:line`. The first seeding used dotted module names and R3 rejected all 11 — correctly, and that is the rule doing its job rather than a nuisance.

Eight mechanisms were **BLOCKED** and then **unblocked**, with the reason logged. Three agents changed nothing (a call that produces no record change produced nothing), one wrote YAML that stopped parsing, one timed out, and two touched `atlas_record.yaml` outside the driver and were restored. Every one was completed on a later call. A stale block is its own kind of lie — `status` keeps reporting work outstanding that is done — so each clearance is written to `_state/unblocked.jsonl` with the evidence that justified it (equations, `state_io`, and a note file all present), rather than quietly deleted.

The excavations are real work, not summaries. `diffusionsolverfe`'s update is transcribed from the OpenCL kernel at L954–L1054; `curvature` is identified as the Menger curvature  $\kappa = 2 \sin \theta / |R - L|$  of three junction-linked centres of mass; `contactlocalflex` is traced through the compiled core to a per-cell `ContactLocalFlexDataContainer` that the identical Python constructor does not reveal.

What the six runs already establish, each with its control (see §5): every CPM mechanism here is an *energy term*, not an update. None of them writes state; they change only the probability that a proposed pixel copy is accepted. That is a structural fact about the whole framework, and it is what Phase 2 will have to normalize against a language whose operators return deltas.

Each mechanism becomes a typed Plexus contract with a verdict, then the skeptic tries to break it.

**The prediction is already on the record** (TRANSFER.md): the *biological* mechanisms — growth, division, death, chemotaxis, secretion, diffusion — should come back largely **alias** or **implementation** of contracts the first atlas already registered, while the *representational* ones — cell-as-domain, the Metropolis acceptance rule, contact energy over shared boundary, volume and surface constraints — should be genuinely new. If that splits as predicted, the conclusion is that *the biological vocabulary is converging while the representational vocabulary is not*.

The measurement, against the same frozen 52-contract baseline the first atlas used:

|                                  | jax-morph | CompuCell3D |
|----------------------------------|-----------|-------------|
| mechanisms scored                | 18        | 24          |
| alias                            | 2         | 4           |
| refinement                       | 2         | 6           |
| implementation                   | 6         | 6           |
| out of scope                     | 6         | 11          |
| <b>genuinely new</b>             | <b>8</b>  | <b>8</b>    |
| yield (new per scored mechanism) | 0.44      | <b>0.33</b> |

The second repository yields **FEWER** new contracts per mechanism than the first — 0.33 against 0.44 — which is the direction the saturation hypothesis predicts and was not guaranteed. It is one data point, not a curve.

**The cross-repository fix earned itself immediately.** Without `--prior`, CompuCell3D scores **9** new contracts and re-counts `adhere` as a first sighting. With it, the four contact plugins (`Contact`, `ContactLocalFlex`, `ContactInternal`, `AdhesionFlex`) are correctly classified as *implementations* of the `adhere` contract jax-morph already proposed. **One contract, double-counted, is a 12% error on the headline** — and it would have grown with every further repository.

**The normalizer found the strongest case itself.** On the last mechanism it reported that CompuCell3D's `SteadyStateDiffusionSolver` solves *the identical equation* as jax-morph's `morphogen` —  $D\nabla^2 c - kc + S = 0$  — and classified it as a second sighting rather than new vocabulary. That is the convergence claim showing up as a concrete match between two unrelated codebases, not as an assertion.

**What is genuinely new here is representational, as predicted.** `occupy` — a cell as a set of lattice sites — is new, and so are `volume_elasticity`, `membrane_tension`, `elongate`, `stiffen`, `bond`, `stay_connected` and `react`. **The prediction in TRANSFER.md was half right:** the biological mechanisms did collapse into contracts the first atlas already had, and the representation did yield something new. But three of the five architectural entries — the Metropolis rule, the energy-sum composition and the MCS time unit — came back *out of scope* rather than new: the normalizer judged them framework mechanics with no biology, exactly as jax-morph's four architectural entries were judged. **Two frameworks, two very different architectures, and both times the architecture scored zero new biological vocabulary.**

**The skeptic earned its place again.** Every new claim was challenged — 14 mechanisms, since the record held more new verdicts than the ledger counts distinct contracts. **Five were refuted and sent back to the normalizer:** `adhesionflex`, `connectivity`, `lengthconstraint`, `surface` and `volume`. `connectivity` was demoted outright from new to `alias`. Nine survived, at confidences of 0.65–0.85.

**And a driver lesson worth recording.** The first attempt ran `step --role normalizer --skeptical` over the already-normalized entries, which re-runs the NORMALIZER first — and a call that changes nothing counts as a failure. Nothing was corrupted (the guard rejected the no-op edits and the measurement was unchanged), but only one challenge actually landed. `--skeptical` belongs *during* the normalizing pass; afterwards, the skeptic must be invoked as its own role.

**Eleven out-of-scope is nearly half the framework**, against six of twenty-four for jax-morph. A Potts framework carries far more machinery that models no biology — trackers, initialisers, dumpers, the acceptance rule itself.

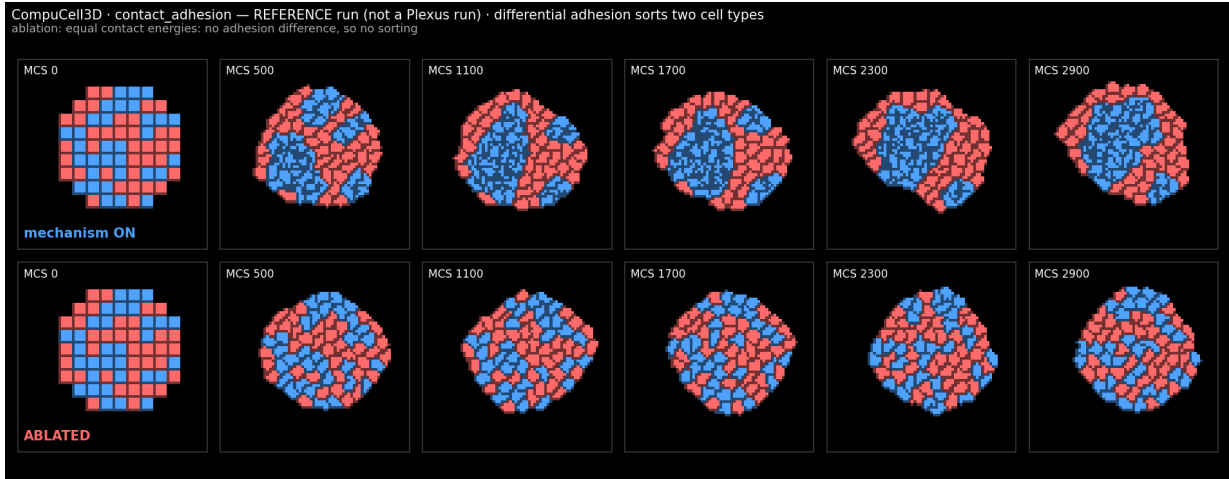
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Phase 3 — Implement what is missing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | not started |
| <p>Every <b>new</b> and every <b>refinement</b> becomes a Plexus operator in the anti-chamber, with a test of one property statable without the reference: a conservation law, a sign, a limit, a symmetry.</p> <p><b>The open question this phase will force.</b> Plexus's cell set is points with per-cell state. A Potts cell is a <i>set of lattice sites</i>. Either that is expressible as an existing set with a different map, or the language needs a representation it does not have — and finding out which is worth more than any individual operator here.</p>                                                                                                                                                                                           |             |
| Phase 4 — Differential validation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | not started |
| <p>Metric and threshold written down <i>first</i>; then the same initial condition through both implementations; then subtract.</p> <p><b>This is the phase most likely to break, and the reason this target was chosen.</b> Every differential test in the first atlas was a subtraction of per-cell arrays. Here there is no per-cell equation to compare and the dynamics are a Metropolis chain, so a matched trajectory is not even meaningful — the same argument that made the division differ compare a pooled <i>hazard</i> instead of cell counts, but now applying to the whole framework rather than one operator. The tests will have to be <b>distributional</b>: equilibrium sorted-boundary length, volume distribution, contact-energy spectrum.</p> |             |
| Phase 5 — The forward figure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | not started |
| <p>One headline result of the paper reproduced inside Plexus out of normalized operators, and set <i>beside</i> the reference rather than described next to it. Cell sorting is the obvious candidate: it is CompuCell3D's canonical demonstration and its outcome is a topology, not a trajectory.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| Phase 6 — Differentiability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | not started |
| <p><b>Probably not applicable, and saying why is the result.</b> The first atlas's Phase 6 rested on the forward model being differentiable. A Metropolis acceptance is a discrete accept/reject on an energy difference — there is no pathwise gradient through it at all, and the straight-through trick that carried the division draw does not obviously transfer. If Plexus can express Potts dynamics forward but not inversely, that is a real and reportable limit of the language.</p>                                                                                                                                                                                                                                                                       |             |
| Phase 7 — Promotion, and the joint ledger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | not started |
| <p>The measurement only becomes an argument here: the cumulative-new curve across <i>two</i> repositories. Flat means the language is saturating; still climbing means it is not.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |

## 5. Six mechanisms, each beside its ablation

Reference runs, not Plexus runs — Phases 2–4 have not started, and every artefact says so on its face. They exist to prove the oracle is real and readable, and to put each mechanism's observable in the currency Phase 4 will need.



| mechanism          | observable                | mechanism ON      | ablated           |
|--------------------|---------------------------|-------------------|-------------------|
| contact_adhesion   | heterotypic boundary      | 290 → 167 (-42%)  | 290 → 388 (+34%)  |
| volume_constraint  | mean volume               | 25 → 59.4 (+138%) | 25 → 0 (-100%)    |
| surface_constraint | mean perimeter            | 20 → 19.3 (-3%)   | 20 → 22.8 (+14%)  |
| chemotaxis         | mean $x$ (source at wall) | 15 → 18 (+20%)    | 15 → 14.8 (-1%)   |
| growth_mitosis     | live cells                | 37 → 296 (+700%)  | 37 → 37 (0%)      |
| external_potential | mean $x$                  | 32 → 57.6 (+80%)  | 32 → 32.2 (+0.5%) |



Cell sorting, mechanism above its ablation on matched frames. Colour is cell type; the seams are cell borders drawn from the id lattice, because colouring by type alone renders a one-type tissue as a solid block and makes the cells — the entire subject of a Potts model — invisible.

**Every ablation separates, and several reverse rather than merely weaken:** sorting becomes mixing, the perimeter grows instead of shrinking, and without a volume constraint the cells do not just relax — they dissolve.

**Every observable is a population or topology statistic, never a trajectory.** That is forced rather than stylistic. A Potts model is a Metropolis chain, so a matched trajectory is not a meaningful object and Phase 4 will have to compare distributions — the same argument that made the first atlas's division differ compare a pooled *hazard* instead of cell counts, now applying to an entire framework.

**Four defects were fixed rather than shipped**, each found by looking at the output: type-level volume parameters silently disabling per-cell growth; type-colouring hiding every cell; a chemotaxis strength that transported cells into the wall and destroyed them; and the `PyCoreSpecs` API not being spelled like `CC3DML`.

## 6. An instrument defect only the second repository could reveal — fixed

`saturation.py` classifies a mechanism implementation when it reaches a contract *this run* has already counted as new. At  $n = 1$  that is complete. At  $n = 2$  it is wrong: every contract the first atlas introduced would be counted as new *again* by the second, inflating precisely the number the instrument exists to protect.

**The fix is a `--prior` flag:** an earlier `atlas_record.yaml` pre-seeds the "already spoken for" set, so a second sighting reads as an implementation *of that repository* rather than as new vocabulary. Checked by feeding `jax-morph`'s record back to itself — its 8 new contracts all demote and it scores

**NEW 0, implementation 14**, which is the correct answer for a repository measured against itself.

Applied to *both* campaigns' copies, so the instrument stays one instrument. That is the whole value of a second target arriving before the first is promoted: **the defect was invisible at  $n = 1$  and structural at  $n = 2$** , and it would have silently doubled the headline number.

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**Where things are.** `TRANSFER.md` — the prediction, written first. `../atlas_jax_morph/STATUS.md` — what the first atlas produced and what was reusable. `_state/registry_baseline.json` — the same frozen 52 contracts, so the two measurements are comparable.